

Spatial Degradation of Hepatocyte Zonation

Using a healthy spatial liver atlas to recover — and watch the collapse of — the lost spatial identity of diseased liver cells



PORTAL

CENTRAL

Combining a healthy spatial atlas (Yakubovsky et al., Nature 2026)
with a single-nucleus disease cohort (Gribben et al., Nature 2024)

Team: [names]

The question, in one breath

A healthy liver is spatially organised: each hepatocyte sits at a position along a blood-flow gradient and does the job that suits its position.

One paper measured that map **with positions kept** (spatial). Another profiled the whole disease course but with a method that **throws positions away** (single-nucleus).

We use the healthy map to put a position back on each diseased cell, then measure how that organisation falls apart as disease worsens.

Healthy lobule



sharp portal-central gradient

Diseased lobule



blurred - identity lost

We turn that blurring into a number per stage.

Liver zonation - and how disease erases it

Zonation. Hepatocytes specialise by position along the porto-central axis: oxygen-rich **periportal** cells run the urea cycle (ASS1, HAL); oxygen-poor **pericentral** cells run detox (CYP2E1) and glutamine synthesis (GLUL).



We measure the first quantitatively, and test whether it travels with the second.

De-zonation

a hepatocyte forgets **where on the axis** it belongs - it co-expresses opposite-end genes.

Epithelial plasticity

a hepatocyte forgets **that it is a hepatocyte at all** - drifting toward a bile-duct (cholangiocyte) identity; biphenotypic cells express both ALB and KRT7/KRT19.

Two papers, mirror-image gaps

Paper 1 - the cells (Vallier 2024)

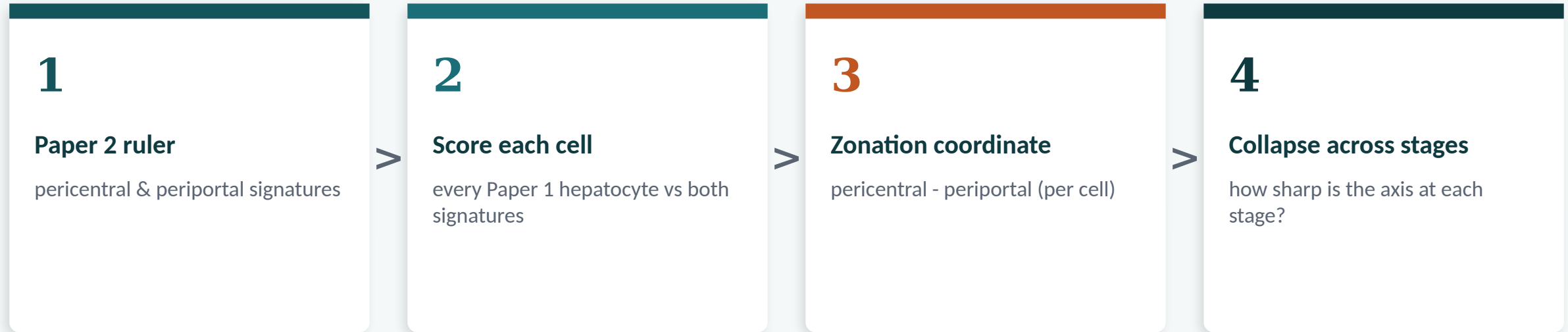
- 47 biopsies, healthy to end-stage
- 69,426 single-nucleus hepatocytes
- Notes zonation loss - only qualitatively
- Weakness: snRNA destroyed the spatial map

Paper 2 - the ruler (Itzkovitz 2026)

- Healthy human liver, spatial transcriptomics
- Quantitative porto-central zonation map
- 1,724 zoned genes - our signatures
- Weakness: healthy only, no disease stages

The gap: one has the map but no disease; the other has the disease but no map. Nobody has used the healthy ruler to decode the disease cohort - that is our project.

Use the healthy map to decode the diseased cells



Honest framing: the reference is a 1-D axis, so we recover a pseudo-zonation coordinate - not a physical (x, y) position.

Three hypotheses, three tests

H1 - erosion

Claim. A de-zonation score rises Healthy to end-stage.

Test. Ordered-trend test (Jonckheere-Terpstra / Spearman on stage rank): does it climb step by step?

H2 - what fails first

Claim. Specific programs (e.g. pericentral CYP detox) collapse earliest.

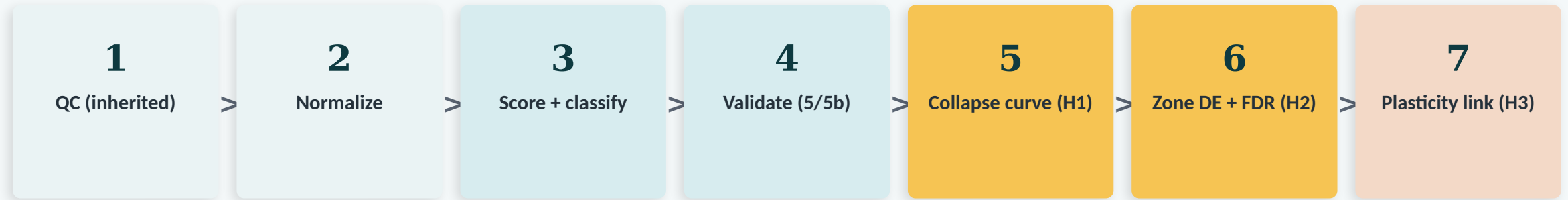
Test. Per-gene differential expression within zones + Benjamini-Hochberg FDR to control false positives.

H3 - plasticity link

Claim. The most de-zonated cells are the ones turning duct-like.

Test. Compare plasticity (KRT7/19) in de-zonated vs zonated cells - Mann-Whitney U.

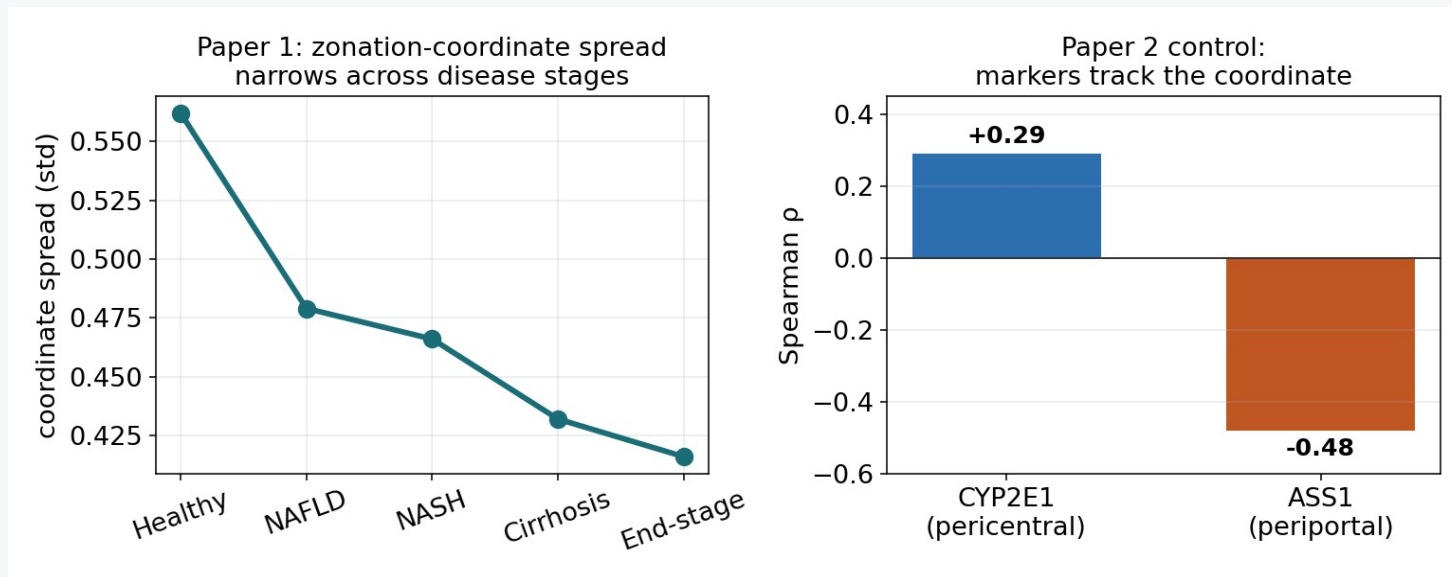
The pipeline - course methods, end to end



All in scope: single-cell processing & clustering (Lec 15-16), differential expression, ordered-trend testing, and Benjamini-Hochberg FDR (Lec 5); counts & complexity from Ex 1-2.

Two-person split: one drives the reference + statistics, the other the disease data + modelling; safe baseline by end of Day 1, rich analysis on Day 2.

Feasibility - the engine already starts



Real Paper 1 hepatocytes: the coordinate's spread narrows monotonically across stages (feasibility run).

69,426

hepatocytes extracted, all 5 stages

rho +0.29 / -0.48

Paper 2 control: CYP2E1 / ASS1 track the axis

loads & runs

both datasets public; pipeline validated

Sanity-check only - the full analysis is the hackathon.

Why it matters - not 'we filled a gap'

1

A reusable decoder

once validated, it reads zonal identity out of the many existing dissociated disease atlases that have no spatial data.

2

What fails first

which zonal programs lose legibility earliest (H2) predicts the order in which liver functions fail.

3

How identity breaks

when/how the program becomes unreadable (silenced vs identity-mixed), tied to the plasticity linked to disease severity (H3).

Not a reproduction: Paper 1 showed zonation loss by eye (2 markers, 2 extremes); we make it quantitative, continuous, transcriptome-wide and stage-resolved.

The ask

**A rough green-light on this direction -
so we can start prep now, not lose the runway.**

Question locked · data downloaded & extracted · pipeline validated on real data · primer written.



Thank you - [team names]